

SWYFT Record Twin, status at day 19 of 26

A right-of-way record map goes in. A checked, georeferenced object record comes out, on the LiDAR, with a queue of what could not be proven. This page says where that stands on 2026-09-29, six days before the demo. **Accuracy now means reconstructability** — how much of the survey can be rebuilt from the record alone, JR's definition, adopted 2026-09-27.

DATE	2026-09-29
DEMO	2026-10-05, 25 min live
BRANCH	caltrans-spike, loop 19 (2026-09-29 13:42)
DEMO SHEET	R-10434.2, Presidio, SF-101
TEST SET	R-10434.1/.2/.3 (Presidio tile) + R-10741.1/.2/.3 (Marin tile) + gap tile (3DEP 2018) · bench also R-71.71, R-10258.1, R-71.70 · 1969 R-65.1..4
RUN	python spike/demo.py --fast · 160 s cold (gate 180 s) · --six 923 s (loop 14; not re-run this loop)

42.6% Drawn boundary rebuilt from record, each foot of ground counted once across six sheets 27,171 of 63,806 ft · was 29.5% at the start of loop 16 (09-27)	55.6% Boundary the record dimensions, rebuilt 26,678 of 47,946 ft · was 38.9%	2 of 19 Parcels rebuilt from record, of the official AREAS tables 61806-9 within 0.1% of record area · 46825-5 via DETAIL "A" · was 0	1.0% Anchored: rebuilt stretches landing on a printed coordinate 660 ft, within ~0.01 ft
0 wrong Gold-keyed pass precision, six sheets 479 passes keyed right, 0 on the wrong line	39 Fails that are genuine record-vs-drawing disagreements e.g. R-10741.1 371.76' printed, 364.61 ft drawn	0.04 ft georeferencing RMS, 18 of 19 control points no hand-typed coordinates	0.17 ft worst offset where adjacent sheets draw the same line fitted independently · shared labels 0.19 ft median

1 drafting error found in the record itself R-10434.1 alignment row L prints northing 2,21,919.67	44/44 table rows read clean from the CAD text 111/111 keyed cells match, R-10434.2	1 strip tunnel easements 61985-1..4 as drawn one envelope, not four figures · settled, not blocked	6 / 6 sheets georeferenced from their own callouts two tiles · package check ≤ 0.4 ft, offset on none
6 / 6 sheets with LiDAR under them third tile from USGS 3DEP (2018) for the gap · 9 sheets on the bench			

Where we are

As of 2026-09-29 (day 19 of 26, six before the demo): the headline is reconstructability, not pass rate — how much of each sheet's drawn boundary a record course (a bearing and distance, a curve with a record direction, or a walk between two printed coordinates) rebuilds on its own, with no drawing shown. Six sheets, each foot counted once: **42.6%** of drawn boundary (27,171 of 63,806 ft), **55.6%** of the boundary the record actually dimensions (26,678 of 47,946 ft), **2 of 19** parcels closed against the official AREAS tables, **1.0%** anchored on a printed coordinate. Four overnight loops (16–19, 09-27 to 09-29) got there and are told loop by loop below; the accuracy section further down has the full per-sheet table, what moved it, and what still limits it.

Two weeks of spikes proved the pieces first. The plan reset on 09-22 named the wall: not reading, but **association**, which drawn line a printed number describes. The 09-23 overnight loop attacked that on the Presidio set with the CAD structure hidden in the PDFs (layers and AutoCAD SHX text annotations) and roughly doubled the pass rates. A second loop the same afternoon made the bench deterministic, tripled the arc-length passes by splitting the long R/W curves where the drawing marks a boundary, and settled the tunnel-easement question: the sheet draws them as one strip, so the four are queued by design, not by failure. That pass-rate-and-coverage framing carried through loop 14 (see the timeline); loop 16 replaced it with reconstructability.

A third loop the same evening (18:00 to 21:10) took the test set from three sheets to six: the Presidio pair plus three sheets on the Marin approach of the Golden Gate

Bridge (R-10741.1/.2/.3, 2017, Route 101, drafter CHaldenwang, a Civil 3D 2012-style export with no SHX annotations and no tables, read by OCR). All six are now georeferenced from their own printed callouts and placed on their tiles; Caltrans' packages check all six and are the offset on none. New tile plumbing (`tiles.py`) routes the LiDAR tail per sheet, `demo.py --six` runs the other five sheets through the chain (640 s, 0 failures), and `qgis_project.py --six` builds a two-tile, six-sheet project. Fixing two state bugs made the baseline reproducible: the closing row is `loop3-final` . Coverage: LiDAR lies under four of the six sheets; the bridge and the strait, where the tiles do not meet, need USGS 3DEP. Leg 5 attributed the north misses from 12 crops: the drafter writes the curve delta as `A=` and OCR joins angle and length into one block, so no curve on the north sheets was ever checked. Both are read now: curve $L=R\Delta$ passes 2 of 2 on R-10741.2 (exact), wrong-line 5 to 3 on .2 and 2 to 1 on .3, no other sheet moved. What is left there has no drawn line to check: chord citations beside curves and a not-to-scale detail inset.

A fourth loop (21:20 to 00:40) went after coverage. A third LiDAR tile from USGS 3DEP (2018, about 10 pts/m²) now lies under R-10741.1 and the east edge of R-10434.3, so every baseline sheet has ground beneath it; R-10434.3 itself turned out to sit 89.5 % on the Presidio tile once its fit was redone, so loop 3's "gap" claim for it was a stale fit. Two more annotation-bearing sheets place from their own control (R-71.71 by grid lines, R-10258.1 by callouts and grid, package agreeing to 0.09 ft), and a third, R-71.70, now takes NGS first-order marks as control: both PIDs fetched, both brass disks found by leader, but the two-point fit disagrees with the sheet's own bearings by 1.29°, so it stays unverified. On the six: the north drafter's dashes were 8.6 to 8.9 pt and the 8 pt cap had been chaining tick marks instead, so the dashed parcel line beside S16°20'26"E now carries its bearing (20/20 on R-10741.2); R-10434.3's line table was masked short by an unrelated curve table at the same x, so 12 table cells had been checked as labels (wrong-line 45 to 33, every pass kept); comma-grouped distances read. One regression caught by the canonical bench and reverted before it shipped: a sliver guard added for R-71.70 had rerouted 462 real annotations on R-10434.3.

A fifth loop (09-24 evening) went after the south bearings, then the arcs. The bearings had one root cause: a chained line kept its first segment's direction and handed it to every piece cut from it, so every label along a long multi-piece record line was compared with one arbitrary heading. Each piece now takes the direction fitted through its own points, and the three south sheets went from 30/49, 18/31, 32/51 to 34/49, 20/31, 37/52 with nothing lost. The arcs did not move, and the attribution says why: the compound-curve story from loop 2 is down to 8 of 65 misses; 32 are bare distances a length-blind fallback was scoring as arcs (now gated, 7 fake fails gone), and 25 are standalone L= labels with no drawn arc within reach, the next thing to look at.

A sixth loop the same night took the six accuracy items in order. Arcs: a lone L= label now sums the contiguous pieces of the one curve it sits on, which brought back the 573.93 (T) total as 52.22 + 20.63 + 501.10 and five more (+6). Bearings: a tolerance that scales with the drawn piece's length, capped at 30 arcminutes, with pieces under 10 pt queued as too short to carry a bearing at all (+10, and five earlier passes on 5 to 8 pt pieces retired as luck); a bearing beside a curve is checked as the chord between the arc's ends (+4); a parallel neighbour is chosen by the printed bearing (+1); at a busy vertex the leader's landing piece wins (+1). Not-to-scale detail bubbles are now found by their dashed outline, caption or no caption: four on R-10434.1, one on R-10741.2, and four passes that sat inside them retired. Two hypotheses died honestly: the north drafter's font is not the read limit (the open simplex and romans strokes are identical to the seed already in use; the unread blocks are prose paragraphs), and R-10434.1's lost closed chain was a transient state that never closed for real. Bearing denominators now count checkable pieces only.

A seventh loop, the next morning, tried the scan idea from gap 1 on R-65.2: cluster the glyph images of one hand, name each cluster once, read the rest by template. The clustering is clean (98.6 % purity on the keyed glyphs once glyphs are separated) and the naming works (57 of 61 agreed boxes vote, the vision model names 36 clusters from their contact strips with no disagreement), but the reader lands at 7 right, 0 wrong, 25 queued on the keyed 32 against the two-reader consensus's 16, 1, 15. As a third vote it lifts the union to 18. What stops it is not recognition: hand digits touch and fuse in 197 of 273 boxes, hatch runs through text, and 8 of the 32 keyed boxes are detector boxes sitting on the wrong text before any reader runs. The next lever on scans is text-line detection on the binarised page.

An eighth loop, one leg, came from looking at one crop: the compound R/W curve on Presidio (R=1470.00', L=573.93'(T)) whose record is the table rows C15 and C16 with no drawn boundary between them. The run-sum rule existed for inline labels and for the small fillets, but the table-tag path grouped tags by the piece their leader landed on, and C15's leader lands on a radial tick. The tag path now uses the same parent-tagged arc pool as the inline checks and searches a tag's own curve for a window whose length matches: C16 passes as its two pieces (521.73 vs 521.76), C19 passes, C19's false radius fail is gone, and C21 is now measured on the right piece, where it still fails for real. Presidio row checks went 31 pass / 12 fail to 33 / 9 with nothing else moving on any sheet. C15 itself cannot pass alone: its drawn piece is C15 plus an untagged C18, and the (T) total that covers them already passes.

A ninth loop, one leg, followed the same crop. The thin alignment curve C21 points at is drawn as several PDF paths end to end, so curve paths whose ends meet with matching tangent and radius are now one parent; dash-dot patterns chain into trains like dashes; and a table's consecutive rows are checked as one run against the drawn

run between the marks that exist, even when a row has no tag on the drawing. That last rule is what C15 needed: it passes as C15 + C16 + C18, the untagged neighbour, 573.95 against 573.93. C20 now reaches a check and fails honestly, since its leader lands on the alignment centreline. C21 is measured rather than fixed: its stitched parent reaches 706.79 of the record's 876.88 ft, and the next path, 10.65 pt away, is a different curve. Two things were tried and reverted with numbers: a wider join gained nothing on C21 and moved a neighbour's radius across tolerance, and dash-dot pieces reached by proximity alone produced six fake fails on R-10434.3, so they now need a leader.

A tenth, short loop traced C21 by eye with the checker's own candidate pool drawn on the sheet. It is all there: a tangent, then five pieces of one thin line across the C9/C10 fillet ($255.91 + 193.93 + 86.86 + 278.14 + 62.04 = 876.88$ ft), ending where three straight lines start along the curve's tangent, a point of tangency. Two rules stop the checker from seeing that: the join across the fillet compares a blended whole-piece radius (2.8 % off), and there is no point-of-tangency cut. Both fixes were tried and rejected on the no-loss gate, with numbers: radius fits on 40 ft windows swing from 1535 to 1382 ft at this sheet's two-decimal precision (100 ft is the floor), and a sheet-wide tangency cut fragments other curves and costs three passes elsewhere. The diff is kept; the scoped version is the next step. Nothing shipped; the loop 9 numbers stand.

An eleventh loop answered a question about the screen: why the purple road surface stops abruptly. Purple was the LiDAR vendor's bridge-deck class alone; it ends where the road leaves the structure, at the Battery tunnels (no return at all), where the class thins at parapets and joints, and on the 2018 3DEP tile, whose pavement came out as 190 fragments under the polygon floor. The class audit found ground on class 2 and deck on class 17 on every tile, so the fix was geometry and assembly, not classes: deck now grows from class-17 seeds into dark, flat cells two metres above the ground DTM; fragmented tiles get a wider closing; and one highway surface per tile dissolves pavement and deck inside the right-of-way faces, marked structure or at grade. The tunnel easements are drawn as record only, below ground. Gains in area are small and honest; the continuity on the screen comes from the union. Found on the way: the one-command demo has slipped to 245 s after the curve work of loops 8 and 9, the first item on the list.

Loops twelve and thirteen (09-26, midday and afternoon) came from a review of what the earlier loops could not see, and the answer was mostly measurement. First, **is a pass on the right line?** Every checked label on two sheets was keyed by eye from a crop: on R-10434.2 and R-10434.3, 286 passes are on the line the drafter meant and none are on a wrong one, with one left unsure; the luck passes found by eye in earlier loops do not recur. Second, **what share of the drawing is checked?** Every value-shaped token on the three Presidio sheets now has a named fate (a table cell, a note,

an area, a radial, a pass, a fail, a queue reason); none drops silently. Over the labels a check could address, 98 of 170 pass on R-10434.2, 60 of 168 on R-10434.1 and 115 of 246 on R-10434.3; that is the honest coverage figure and it is on the bench now. Third, **independent checks**: Caltrans' printed alignment coordinates land on the fit at 0.02 to 0.07 ft; $L = R \cdot \Delta$ holds on all 68 curve rows; adjacent sheets, each fitted from its own callouts, draw the same line within 0.17 ft. Those checks found a drafting error in the record itself (R-10434.1, alignment row L, northing printed 2,21,919.67) and a reader bug that had dropped twelve distances from an R-10434.3 table (fixed). Coverage work then resolved tags with an unread glyph from the CAD text or from the one table row that passes its own check, and checked bare (T) run totals with the printed bearing required on the run: 16 new passes, none lost, every new crop viewed (one wrong-line pass was caught that way and the rule tightened). A fourteenth loop cleaned up the fails those new paths had brought in: a result far beyond any keyed real disagreement is now queued as “wrong line likely”, so of the 16 fails left on the demo sheet, 13 are real record disagreements. The demo run is under its gate at 160 s cold. What could not be shown: none of the six sheets draws buildings or pavement edges, so the 0.668 m epoch shift cannot be tested on the ground from these sheets.

Overnight 09-27, a scan detector bake-off on the 1969 R-65.2 sheet (outside the six-sheet set) found PP-OCRv5's server detector lands 30–32 of 32 keyed boxes against Surya's 24 (which fuses $R=\Delta/L$ stacks) and a vision model's 1; wired in opt-in behind the existing RapidOCR default. Not part of the reconstructability story below, but it is what will eventually unblock the scans.

A fifteenth loop (09-27) fixed how a fail earns its keep. An arc fail now stands only when its association is proven — a lone candidate in the leader's reach, or a clear touch — otherwise it queues as unproven; that alone cut R-10434.3's fails 44 to 27 and its keyed-wrong count 24 to 7, all 19 real fails kept. A second cause of unfound labels turned out to be a box-rotation prefilter throwing out the right line whenever text was stacked; letting a bearing-agreeing candidate in added 12 passes, every one viewed. A gold set on R-10434.1 (83 right, 0 wrong, 7 real fails, 9 wrong-line, 4 unsure) joined Presidio's and R-10434.3's, so every keyed sheet read zero wrong-line passes.

Loop 16 (09-27 23:05 to 09-28), JR's ruling: accuracy means reconstructable. A foot of drawn boundary counts only when a record course walks it, unflagged, within 0.5 ft; a parcel counts only when every edge is walked by record and the figure closes within 1 ft — nothing about whether a label merely sits near a line. Measured cold, the six sheets rebuilt 29.5% of drawn boundary (20,876 of 70,661 ft), 38.9% of what the record actually dimensions, and 0 of 224 named parcels. Building the metric surfaced a triple-counting bug on the way: near-coincident copies of the same stroke (an R/W line drawn on top of a face edge) were being credited up to three times over; fixing it is why these numbers, and loop 17's, read lower than a naive read of the underlying

passes would suggest, not higher. A ceiling measurement (leg M) found the honest limit: of the boundary these six sheets dimension anywhere, about 70% is reachable in principle; the rest is bearing-only, curves without a recorded direction, or already in honest disagreement with the drawing.

Loop 17 (09-28) found the single biggest lever. The R-10741 sheets' bearings were being read fine all along; what silently dropped them was text canonicalisation — OCR renders the compass letters lowercase and the degree/minute/second marks as ' * " instead of °", and the parser threw the whole bearing away instead of normalising it. Fixing that alone was the loop's biggest single gain, about five points of the set. Alongside it: traverse misfit moved from a cumulative judgement (one bad edge failed every edge chained after it) to one judged per edge, curve radii earlier passes had computed and then discarded now reach the labels that need them, and refused tags stopped leaking into the traverse as if they had passed. Set headline: 38.1% of drawn boundary, 49.0% of dimensioned boundary, first parcel closed (Presidio 61806-9).

Loop 18 (09-28 evening) taught the walk to use the record's own coordinate math. Per JR's ruling, a line between two printed coordinates is now a record course in its own right (an inverse), a "(T)" total counts for every piece of the one straight or curved line it labels, and a radial at a curve end counts only when the record circle it implies passes within 0.5 ft of every drawn point of that curve. R-10741.1's L=660.20' curve had been refused by a direction check that treated its two consistent end tangents as disagreeing (they differ by exactly 180°), and later lost the radius printed on its own label; both fixed. Curves gained two rules: two that happen to share a fillet no longer merge into one, and a curve that fails its own internal check is no longer walked as if it had passed.

Loop 19 (09-29), the coordinate-anchored rebuild queued since T-001. Starting at a printed coordinate on the boundary, the walker applies the record's own courses in order and accepts a chain only when it lands on the next printed coordinate within tolerance — a closure failure names the likely course, not just a mismatch. It closes: Presidio's four-course walk from one printed coordinate to the next lands 0.007 ft off (figure below). Anchored coverage is still small (1.0%, 660 ft) — few stretches of boundary run between two printed coordinates with nothing missing in between — but where it applies it is the strongest proof on the page: no drawing measured, only the record's own numbers walked to a value the drafter also printed. Cross-sheet chaining and closure-checked fills came back with nothing to show (0 joinable, 0 checkable) except one real find: DETAIL "A" on R-10741.2 closes parcel 46825-5, the second parcel of two. New gold sets on all three R-10741 sheets keyed every checked label; combined with Presidio's and R-10434's, that is 479 passes checked by eye across six sheets, 0 on the wrong line.

Tried and rejected the same day: loosening tolerance. JR asked what a full foot of slack (distance, bearing as a lateral offset, misfit, anchoring) would buy. Measured: +0.4 points of set boundary, all on R-10434.3, for 4 new wrong-line passes on the gold sets and 3 hidden real record disagreements. Not worth it; the defaults stand (distance 0.30 ft + 0.05%, bearing 0.05°, 0.5 ft rebuild bar).

09-09 to 09-21

Spikes

Text from strokes, OCR, georef from callouts and grid ticks, LiDAR Q1-Q5, scan census. 8 of 11 CAD sheets georeference.

09-22

Plan reset

Association is the wall. Product = the numbers walked as a traverse. Scans = QA story, not input. 13-day roadmap.

09-22

demo.py, objects, QGIS

One command, offline. Object record with provenance. Exception page. Figures and numbers from the run.

09-23 04:00-09:00

Overnight loop

SHX annotation reads, association filters as defaults, traverse edges paired, 1969 scans on the tile. 7 commits, every gate re-measured.

09-23 12:00-14:30

Loop 2

Twin annotations folded. Dash trains on the easement layer. Compound curves split. Arcs 4/11 to 13/19. Easement strip measured as one envelope.

09-23 18:00-21:10

Loop 3

Six zones as the baseline: north tile plumbed, R-10741.1/.2/.3 georeferenced from their own callouts, two-tile QGIS project, bench row loop3-final; north curve data read, wrong-line down.

09-23 21:20-00:40

Loop 4

Coverage: 3DEP gap tile, ground under all six; R-71.71 and R-10258.1 placed; NGS PID control built (R-71.70 unverified, 1.29°); north dashes seen, R-10434.3 table cells masked; bench row loop4-final.

09-24 18:00-19:15

Loop 5

South bearings +11 from one cause (cut pieces inherited the seed direction); arcs re-attributed: fake arc fails gated out, fillets matched by radius; bench row loop5-final.

09-24 19:30-23:45

Loop 6

Lone L= arcs summed on their curve (+6); bearing tolerance by piece length, chords beside curves, parallels by printed bearing (+16); detail bubbles by geometry; font-seed and closure hypotheses falsified; bench row loop6-final.

09-25 08:45-09:25

Loop 7

The scan by its own alphabet on R-65.2: clean clusters, thin coverage; a third reader (union 18 of 32, 0 wrong); detector boxes on the wrong text are the larger loss.

09-25 11:10–17:10

Loop 8

Curve table tags on the shared parent-tagged arc pool; Presidio row checks 31/12 to 33/9; C15 shown to be a C15+C18 piece; bench row loop8-final.

09-25 17:40–20:15

Loop 9

Curve paths stitched into one parent; dash-dot trains; consecutive table rows as a run (C15 passes); C21 measured, not fixed; bench row loop9-final.

09-26 00:40–03:10

Loop 10

C21 traced by eye: whole on the sheet, five pieces to a point of tangency; two fixes measured and rejected; nothing shipped.

09-26 03:30–04:40

Loop 11

The road as one surface: class audit per tile, deck by height above ground, gap-tile pavement, tunnels record-only; demo --fast regressed to 245 s.

09-26 11:30–13:00

Loop 12

Review gaps: demo 245 to 160 s; gold set on R-10434.2 (132/132 right); alignment tables and $L = R \cdot \Delta$ as self-checks (a record error found); matchlines within 0.17 ft; table reader fixed.

09-26 13:00–15:15

Loop 13

Gold set on R-10434.3 (0 wrong); every token attributed, honest coverage on the bench; unread tags resolved by the record, (T) totals checked: +16 passes, 0 lost.

09-26 16:30–17:30

Loop 14

Wrong-line fails queued instead of reported: fails 26 to 16 on the demo sheet, every real disagreement kept, passes unchanged.

09-27

Scan detector

PP-OCRv5 server detector 30–32/32 keyed boxes on the 1969 scan vs Surya 24, qwen 1; wired in opt-in.

09-27

Loop 15

A fail needs proof: R-10434.3 fails 44 to 27, wrong-line 24 to 7. Box-rotation cause found (+12 passes). Gold set on R-10434.1.

09-27 23:05–09-28

Loop 16: accuracy redefined

JR: accuracy = reconstructable from the record. Metric built, a triple-count bug caught. Baseline 29.5% / 38.9% / 0 parcels; ceiling ~70%.

09-28

Loop 17

OCR bearing symbols fixed (biggest single gain), misfit judged per edge, discarded curve radii recovered. 38.1% / 49.0%, first parcel.

09-28 evening

Loop 18: record anchors

Printed-coordinate inverses, “(T)” totals as courses, radials checked against fitted curve centres.

09-29

09-29

Loop 19: anchored rebuild

Record walked between printed coordinates, closes to 0.007 ft; DETAIL “A” closes 46825-5; R-10741 gold sets. 42.6% / 55.6% / 2 parcels.

09-29 now

Demo prep

Slides, Q&A sheet and recording not started. Freeze 10-03.

Tolerance experiment

1 ft of slack tried and rejected: +0.4 pt for 4 new wrong-line passes, 3 hidden record disagreements.

to 10-05

Freeze, rehearse

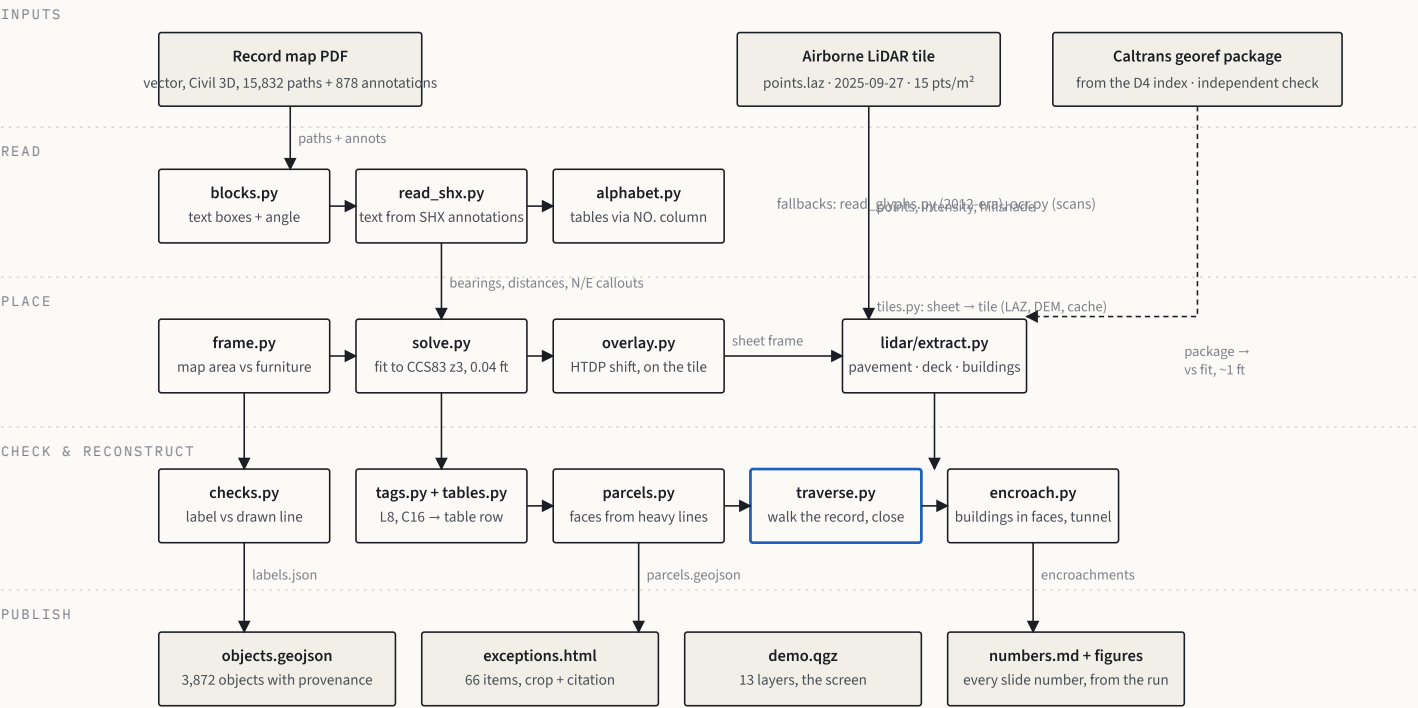
A coordinate-table reader for bearing-only lines. Fallback recording. Two rehearsals. Freeze by 10-03.

Roadmap items

DAYS	DELIVERABLE	STATE	PROOF
1-2	One command, offline, under 3 min. Object record with provenance. QGIS project.	DONE	122 s, 21 steps ok; 3,872 objects each with sheet, rule, residual, status; 13 QGIS layers valid, render PNG as proof
3-6	Table traverse: tags on the drawing tied to table rows, figures walked and closed	DONE AS FAR AS THE SHEET ALLOWS	tags associated 24 of 44 rows, line checks pass; 2 chains close on the demo sheet; the easements are one drawn strip, measured and queued as such (below)
7-8	Exception page a surveyor can work	DONE	one offline HTML, 66 items in cause groups, crop + citation + resolution field per item; scan's queue on the same page
9-10	LiDAR numbers and figures refreshed from the run	DONE	numbers.md and four figures regenerate at the end of every run; tunnel profile caveated until easements close
11	Slides for the non-demo minutes, Q&A sheet, recorded fallback	NOT STARTED	Q&A answers drafted in ROADMAP.md; no recording yet
12-13	Two rehearsals, freeze	NOT STARTED	—

The demo workflow

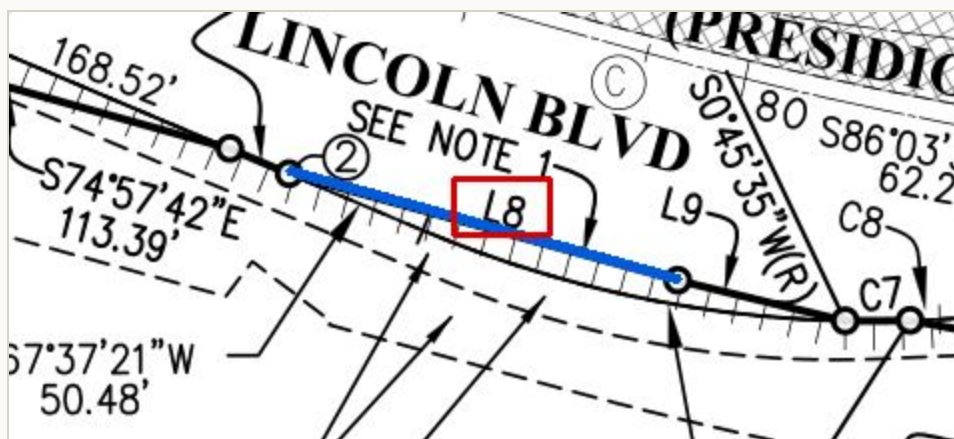
Everything below runs from `python spike/demo.py --fast` with no network, 129 s. A `--six` flag adds the other five sheets' tails and the six-sheet QGIS project, 640 s, 0 failures. Twenty-one scripts in fixed order, each reading the files the previous ones wrote under `spike/out/`. Read once, then every downstream number is computed from geometry, never re-read.



The pipeline as it runs. Text is read once from the CAD annotations, then every check, closure and area is computed from geometry. The traverse step is the demo's thesis: the printed record walked and closed with no label-to-line heuristic. Caltrans' own package enters only as an independent check on the fit.

The five demo segments

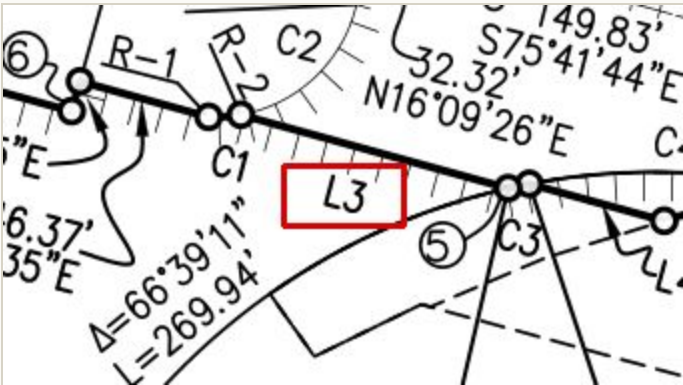
#	SEGMENT	WHAT THE ROOM SEES	STATE
1	Sheet in, objects out	Tables and callouts read; fit residuals; linework on the LiDAR intensity image	LIVE



1. Find the tag on the drawing. Tag text from the SHX annotation. The leader's arrowhead picks the segment (blue): a whole piece between vertex circles.

LINE DATA TABLE		
NO.	BEARING	DISTANCE
L1	N18°06'50"E(R)	1.86'
L2	N14°18'16"E	3.76'
L3	S75°11'28"E	96.27'
L4	S75°12'02"E	48.55'
L5	N72°08'15"E	19.34'
L6	S70°11'40"E	6.29'
L7	S74°42'15"E	39.52'
L8	S75°11'28"E	139.90'
L9	S76°56'20"E	59.89'

2. **Read the row.** The NO. column is a known sequence, so the table validates itself. L8 says S75°11'28"E, 139.90 ft.



3. **Or queue it.** L3 has no leader and two candidates. Ambiguity goes to the queue, never a guess.

READ_SHX.JSON · ONE BLOCK PER ANNOTATION

```
{ "cx": 1837.5, "cy": 757.0, "w": 36.0, "h": 7.26,
  "angle": -12.76, "text": "TIE BACK",
  "conf": 1.0, "src": "shx", "id": 2 }
```

TABLES.JSON · L8 AFTER ALPHABET.PY

L8	S75°11'28"E	139.90'
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TAGS_CHECKS.CSV · VERDICT PER ROW

tag	check	printed	drawn	diff	result	assoc
L3	bearing	S75°11'28"E	S75°10'36"E	0.9'	pass	beside
L3	distance	96.27	96.27	-0.00	pass	beside
L4	bearing	S75°12'02"E	S75°12'43"E	0.7'	pass	leader
L4	distance	48.55	48.56	+0.01	pass	leader

GEOREF.JSON · THE FIT EVERY CHECK LIVES IN

```
"crs": "EPSG:2227", "scale_ft_per_pt": 1.38885,
"rotation_deg": 0.536, "rms_ft": 0.041,
"control": [ { "N": 2119812.80, "E": 5993809.52,
               "residual_ft": 0.053, "used": true }, ... ]
```

 Twelve random passing distance labels, each with the label in a red box and the measured line in blue

Output: twelve random passing distance labels, label region in red, the line the reader measured in blue. This is the gate evidence the overnight loop used: a pass only counts if a human sees it on the right line. All twelve are.

Then the record walks itself

TRAVERSE.JSON · ONE CHAIN, CLOSED

```
{ "closed": true, "n_edges": 5, "full_record": 5,
  "misfit_end_ft": 0.14,
  "edges": [
    { "edge": "N67°07'01\"W + 138.21'", "misfit_ft": 0.07},
    { "edge": "L12", "misfit_ft": 0.11},
    { "edge": "L11", "misfit_ft": 0.07},
    { "edge": "N70°37'16\"W + 168.12'", "misfit_ft": 0.10},
    { "edge": "L13", ...}] }
record area 6,970.6 sq ft · drawn face 6,974 · +0.06 %
```



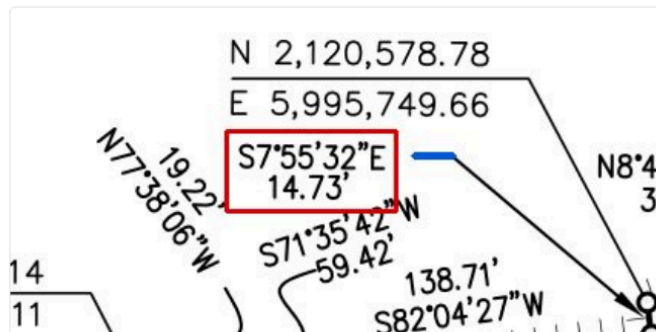
OBJECTS.GEOJSON · ONE OF 3,872 PUBLISHED OBJECTS

```
{ "kind": "control point", "sheet": "r_10434_002_2020-09-16",
  "printed": { "E": 5993809.52, "N": 2119812.80},
  "measured": { "sheet_pt": [797.8, 1026.4] },
  "residual": 0.053, "units": "ft", "status": "verified",
  "rule": "printed N/E callout, leader traced to point,
    similarity fit by consensus; residual vs fit",
  "adjudicated_by": "", "adjudicated_on": "" }
geometry: Point [-122.46557, 37.80025]
```

Every published object carries the sheet, the printed value, the measured value, the residual, the rule that produced it, and an empty adjudication slot. In QGIS a click shows all of it. That is the "reviewer reaches provenance in seconds" promise.

▼ Printed value does not match the line beside it 7

A dimension or bearing disagrees with the line the reader measured (blue) by a small amount. Decide which is the record.



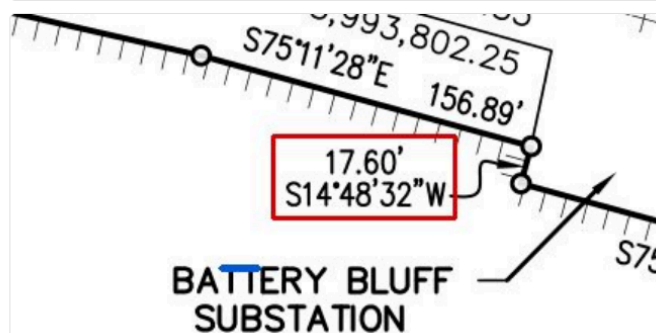
distance

distance: off by +2.60 ft

printed 14.73' · drawn 17.33 ft · +2.60 ft

r_10434_002_2020-09-16 · region 2063, 432, 2118, 458 pt

— resolution — initials / note



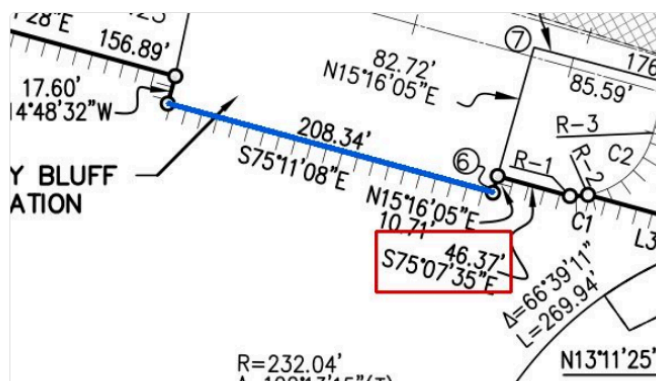
distance

distance: off by -1.35 ft

printed 17.60' · drawn 16.25 ft · -1.35 ft

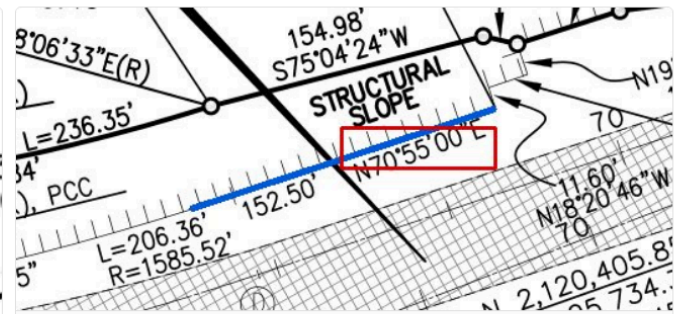
r_10434_002_2020-09-16 · region 712, 565, 772, 592 pt

— resolution — initials / note



bearing

bearing: off by +3.4'



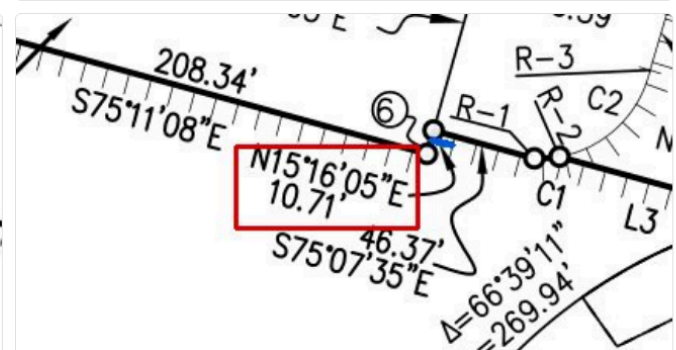
bearing

bearing: off by +18.6'

printed N70°55'00"E · drawn N71°13'36"E · +18.6'

r_10434_002_2020-09-16 · region 1992, 558, 2053, 574 pt

— resolution — initials / note



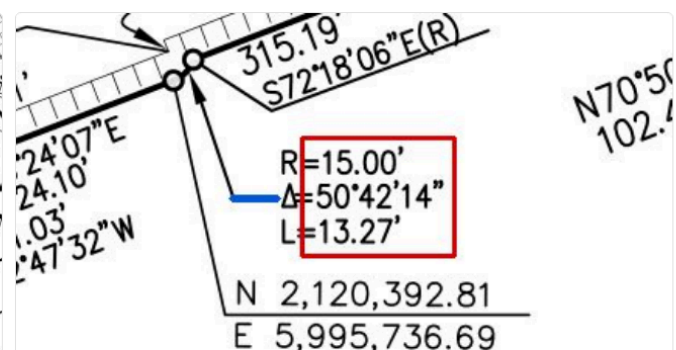
distance

distance: off by -0.75 ft

printed 10.71' · drawn 9.96 ft · -0.75 ft

r_10434_002_2020-09-16 · region 877, 618, 935, 644 pt

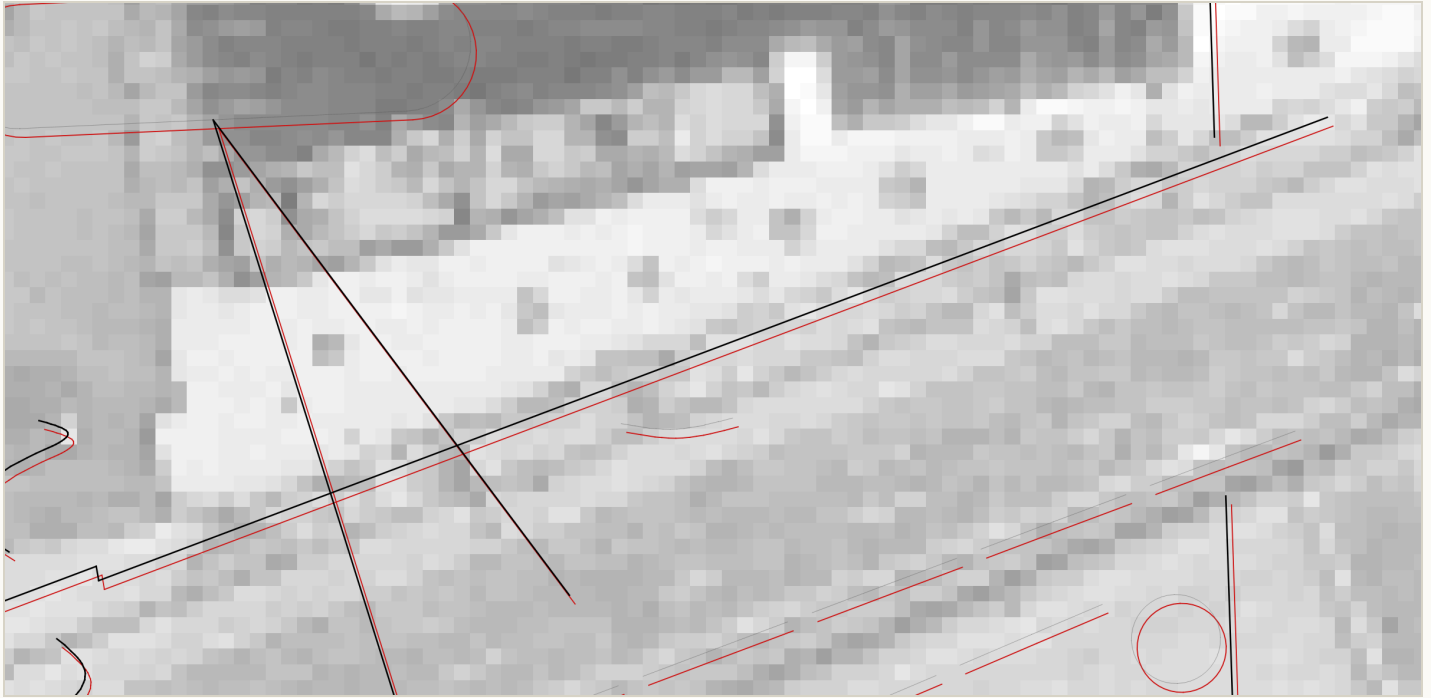
— resolution — initials / note



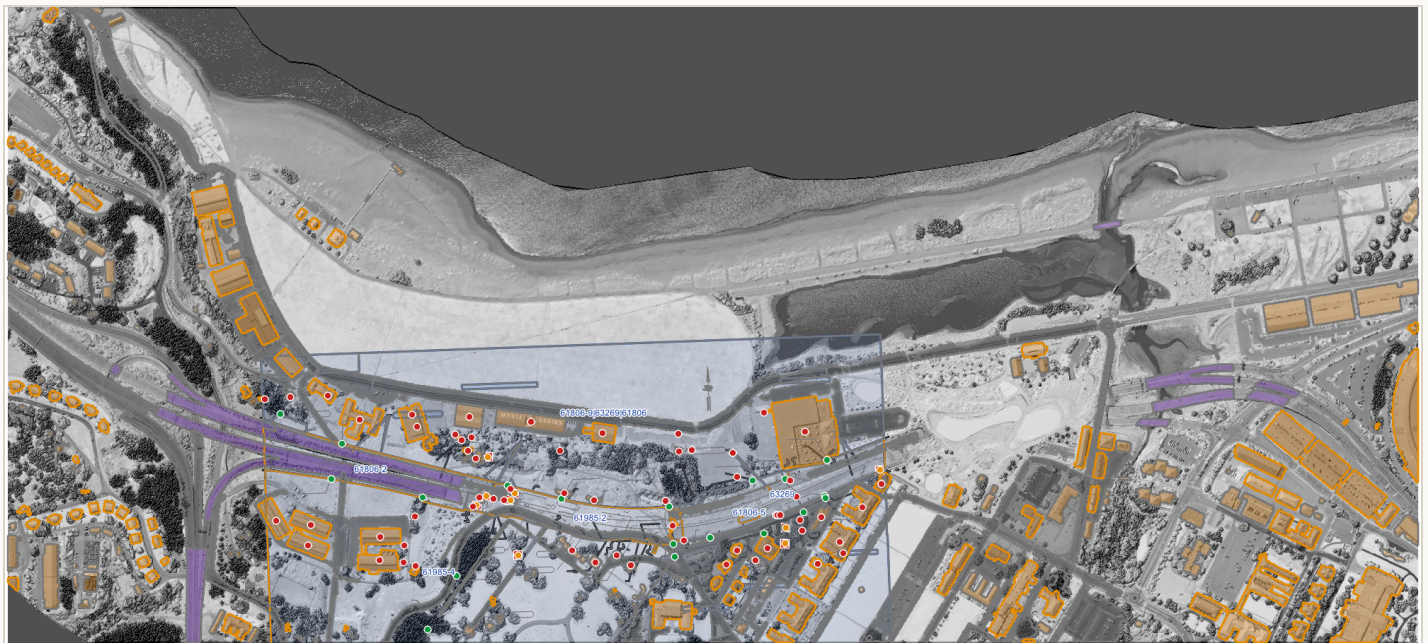
distance

distance: off by +4.58 ft

Output: exceptions.html. One offline file. Each item: crop, cause group, printed vs drawn, citation in PDF points, a resolution a surveyor fills in and exports.



Output: linework on the LiDAR, before and after HTDP. Sheet epoch 1991.35 to LiDAR 2010.0:
0.668 m toward N33°W, applied and shown, never silent.



Output: demo.qgz, the screen. Thirteen layers built by PyQGIS in legend order. The script re-reads the written project, asserts every layer survives, and renders this PNG as proof.

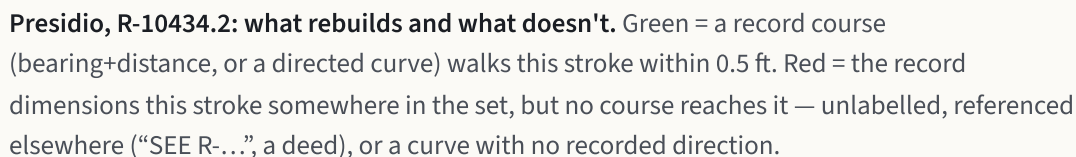
How much of the survey is reconstructable

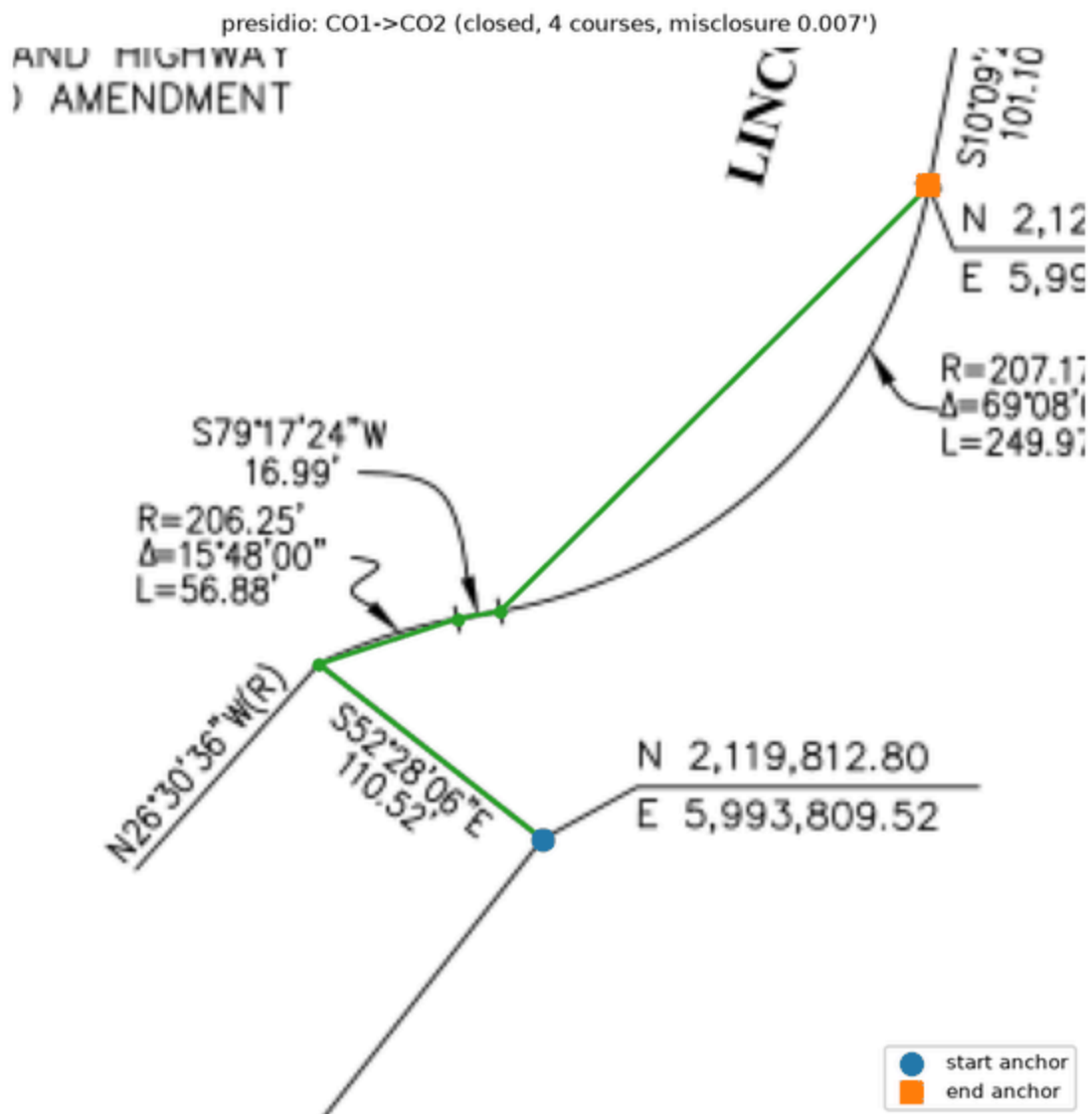
JR's definition, adopted 2026-09-27: a foot of drawn boundary counts only when a record course — a bearing and distance, or a curve with a record chord direction — walks it, unflagged, within 0.5 ft of the drawing; a parcel counts only when every edge is walked by record and the figure closes within 1 ft. Nothing about whether a label happens to sit near a line. Six sheets, each foot of ground counted once (where two R-10741 sheets draw the same ground, it is counted once): **42.6%** of drawn boundary rebuilt from record alone (27,171 of 63,806 ft), **55.6%** of the boundary the record actually dimensions (26,678 of 47,946 ft), **2 of 19** parcels closed against the official AREAS tables (61806-9, within 0.1% of the record area; 46825-5 via DETAIL "A"), and **1.0%** (660 ft) anchored — rebuilt stretches that also land on a printed coordinate within about 0.01 ft.

HISTORY: AT THE START OF LOOP 16 (2026-09-27) THIS MEASURE READ 29.5% DRAWN / 38.9% DIMENSIONED / 0 PARCELS. THAT EARLY FIGURE SUMMED EACH SHEET'S BOUNDARY SEPARATELY AND DOUBLE-COUNTED THE R-10741 TILE OVERLAP; ON THAT SAME (UNCORRECTED) BASIS, TODAY'S NUMBER WOULD READ 43.2%.

SHEET	REBUILT FT	DRAWN FT	%	LABELS CHECKED	WRONG- LINE	GOLD: RIGHT / WRONG	GOLD FAILS: REAL / WRONG-LINE
Presidio	5,531	10,648	52%	102/171	5	134/0	12/1
R-10434.1	2,726	9,079	30%	62/171	7	83/0	5/5
R-10434.3	4,449	17,271	26%	121/248	12	152/0	18/5
R-10741.1	5,957	10,422	57%	32/39	2	33/0	4/1
R-10741.2	6,653	12,416	54%	40/58	0	40/0	0/1
R-10741.3	5,013	10,425	48%	37/48	0	37/0	0/0

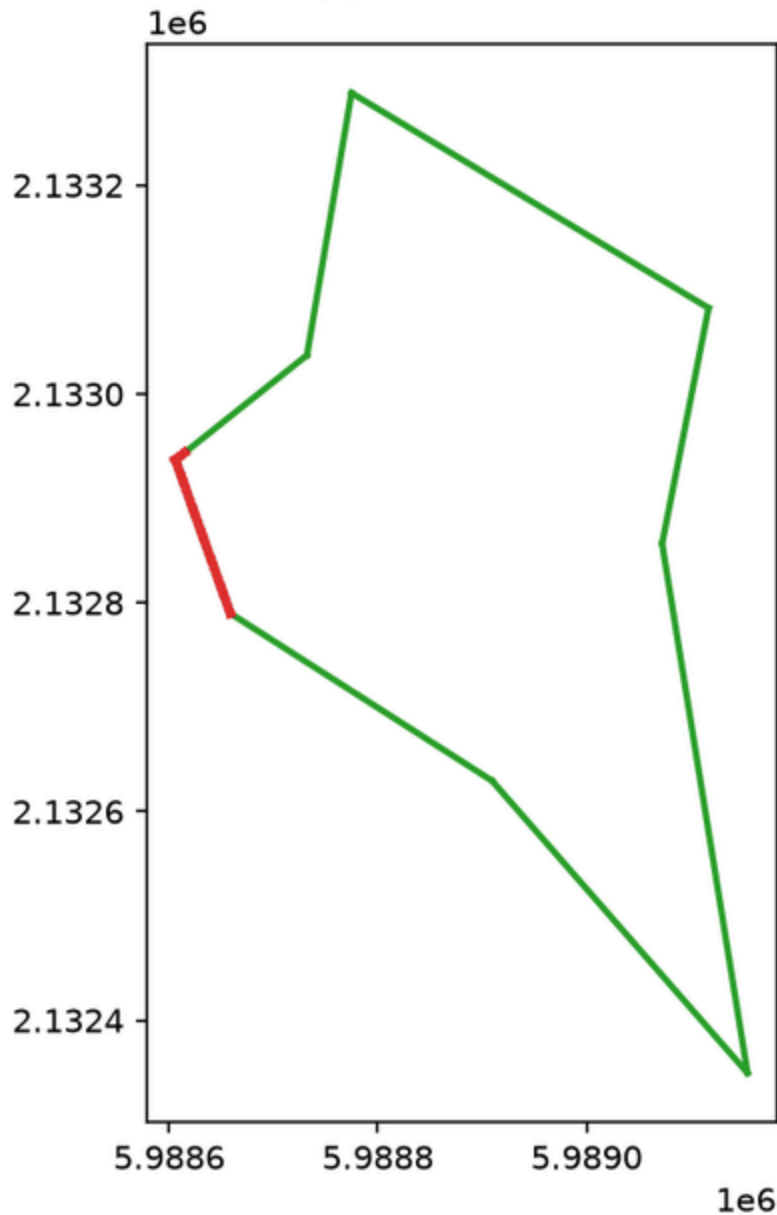
Zero wrong-line gold passes on all six gold-keyed sheets (479 passes keyed right in total). The R-10741 gold sets are new this loop (loop 19 leg 5) — those three sheets hold about 62% of the rebuilt footage, so they were the biggest hole in the precision claim until now. The 39 fails left across the six sheets are genuine record-vs-drawing disagreements, not reading or association mistakes: for example R-10741.1 prints S10°30'18"E 371.76' where the sheet draws 364.61 ft.





A record walk landing on a printed coordinate. Four courses (two lines, two curves) chained from the printed coordinate at N 2,119,812.80 / E 5,993,809.52 close on the next one 0.007 ft off — no drawing measured, only the record's own numbers.

46825-5 (R-10741.2): DETAIL "A" gap 167.09' vs courses 166.97' -- attached, closes=True



DETAIL "A" closes parcel 46825-5. R-10741.2's detail inset supplies the one course (red, 167.09 ft printed vs 166.97 ft summed from the main sheet) the rest of the record leaves out; with it, the parcel closes.

What moved it (loops 16–19)

Measurement made honest first. Each line is now judged on its own, where one bad line used to fail every correct line chained after it on the same run. Duplicate strokes (an R/W line drawn on top of a face edge) are now counted once — they were being credited up to 3×. Non-boundary linework is out of the denominator. The overlap

between the R-10741 sheets (where .1/.2 and .2/.3 draw the same ground) is counted once across the set, not once per sheet. Parcels are scored against the official AREAS tables, not against every named face on the drawing.

Then reading and association. The single biggest gain: the north sheets' OCR renders bearings with lowercase compass letters and ' * ' in place of the degree mark, and the parser was silently dropping the whole bearing rather than normalising it — worth about +5 points alone once fixed. Also: labels on stubs now resolve; refused table rows that had been leaking into the traverse are excluded; curve radii that earlier passes computed and then discarded now reach the labels that need them; radial capture rose from 43 to 98 curve ends; curves that had wrongly merged across a shared fillet split back apart; a curve that fails its own check is no longer walked as if it had passed; R-10741.1's L=660 .20 ' curve is rebuilt again (a 180° direction-check bug, then its own printed radius bound to the wrong piece).

Then record anchors. A “(T)” total now counts as a whole course for every piece of the one straight or curved line it labels (860.77 ft(T) is one course, not several unlabelled pieces). DETAIL “A” closes 46825-5. And walking the record between two printed coordinates, where it applies: 8 chains close within about 0.01 ft, none fail.

What limits it

Share of the dimensioned boundary still not rebuilt, by cause: **62%** has no record course reaching the line at all (52 points of that is unlabelled — much of it only referenced elsewhere, “SEE R-...” or a deed, or never dimensioned anywhere in this set); **17%** is bearing-only lines (the distance is missing); **12%** is curves with no recorded direction; **5%** is distance-only lines (the bearing is missing); **4%** is a genuine record/drawing disagreement. The ceiling, measured at the start of loop 18 with duplicates removed: about **85%** of the dimensioned boundary is reachable in principle. That is the honest limit of these six sheets, not of the method.

Tried and not kept

Measured and set aside: a stronger OCR detector on R-10741 scored 6 of 14 keyed boxes against RapidOCR's 14 of 14, so no change; PDF draw order, grouping and label-placement rules never beat nearest-line association; curve-to-curve chaining completed no curve; coordinate inverses between labelled points, 0 qualify; cross-sheet chains, 0 joinable; closure-checked fills, 0 checkable. The coordinate-anchored rebuild (loop 19, “step four”) works as an independent check but adds little coverage on its own — few stretches run between two printed coordinates with nothing missing in between.

Rejected on the gate: JR asked what loosening every tolerance to a full foot would buy (distance, bearing as a lateral offset, misfit, anchoring). Measured: +0.4 points of set boundary, all on R-10434.3, for 4 new wrong-line gold passes and 3 hidden real record disagreements. Not worth it — the defaults stand: distance 0.30 ft + 0.05%, bearing 0.05°, 0.5 ft rebuild bar.

Next: a coordinate-table value reader for bearing-only lines (the distance is on the sheet, just not beside the bearing); curves still missing a recorded direction; a general detail-inset reader (DETAIL “A” was read by hand this loop); bring the 1969 scans into the score.

Full scorecard, R-10434.2 (demo sheet)

MEASURE	RESULT	NOTE
Reading		
Table rows read clean from SHX annotations	44 / 44	R-10434.1 25/25, R-10434.3 83/83. Hand-keyed cells 111/111 match
Text blocks with exact text	832	annotation text, confidence 1.0; no OCR on this sheet class
Georeferencing		
Control points used, RMS	18 / 19 · 0.04 ft	derived scale 1.38885 ft/pt vs plot scale 1.38889
Agreement with Caltrans' own package	~0.6–1.1 ft	raster-limited; the independent check
Printed vs drawn (labels on the drawing)		
Distance	41 / 49	passes agree to 0.03 ft median, 0.24 max
Bearing	41 / 43	passes agree to 0.55' median
Arc length	16 / 19	was 4/11. Long R/W curves now split at circles, meeting lines and radial ticks; (T) totals checked. Misses left: boundaries the drawing does not mark
Curve L = R·Δ (internal consistency)	10 / 10	
Table rows through their drawing tags		

Rows found on the drawing / associated	33 of 44 / 24	18 by leader arrowhead; 12 partial reads and 33 ambiguities queued, none guessed
Row checks pass · fail	34 · 10	where associated, lines are exact; the arc fails are runs whose sub-arc boundary is not drawn
Reconstruction		
Traverse edges placed / with full record	87 / 49	was 81 / 22 before the loop
Closed chains	2	5 edges, misfit 0.14 ft, area 6,970.6 vs drawn 6,974 sq ft; a second 4-edge loop closes to 0.18 ft
Parcel faces named / within 1 % of the parcel table	8 of 60 / 1	figures leave at matchlines; easement strips do not close
Tunnel easements 61985-1..4	1 strip · 48,410 sq ft	drawn as one envelope; table sum 70,690 sq ft, the TCEs overlap the tunnel easements. QUEUED BY DESIGN
LiDAR		
Features extracted	35 · 11 · 281	pavement, viaduct deck, building polygons (175 pass the roof test); deck now also grown by height above ground (+45 m ² inside the R/W)
Highway surface, Presidio tile	33,988 · 146,620 m ²	structure · at grade, one dissolved surface inside the R/W faces; the blank middle is the Battery tunnels, drawn record-only
Highway pavement inside drawn R/W faces	81 %	9,014 m ² outside within 15 m flagged for review
Epoch shift 1991.35 → 2010.0	0.668 m N33°W	NGS HTDP v3.6.0, real call
Beyond the demo set		
Six-sheet baseline (all six sheets, two tiles)	↑ table above	rebuilt ft, labels checked, wrong-line, gold pass/fail per sheet — see “How much of the survey is reconstructable”, above
Vector sheets georeferenced (all D4 vector sheets on disk)	7 / 17	the 10 refused all fail on reads: other drafters' fonts, no tables to validate an alphabet (before the R-10741 set)

Scan R-65.2 (1969, hand-lettered), 32 keyed boxes

18 right · 0 wrong (gated) · 14 queued

three readers: RapidOCR, a local vision model, and the sheet's own alphabet (loop 7); 8 of the 14 are detector boxes on the wrong text. 48 of 65 sheets in the archive are scans

What the numbers do not claim. The 0.04 ft residual is against numbers read off the same sheet. Independent check is Caltrans' package at about a foot, raster-limited. We claim records-research grade, not a survey tolerance, until a pilot sets one.

Independent checks

HISTORY: PRECISION AND COVERAGE (PASS RATE, WRONG-LINE COUNT, LABELS-CHECKED FRACTION) WERE THIS PAGE'S HEADLINE ACCURACY MEASURE THROUGH VERSION 15. RECONSTRUCTABILITY, ABOVE, REPLACED IT AS THE HEADLINE ON 2026-09-27; THE PER-SHEET GOLD-PASS AND COVERAGE NUMBERS NOW LIVE IN THE RECONSTRUCTABILITY TABLE ABOVE.

Checks added in loop 12 that use none of the labels being checked — against the record itself, and against neighbouring sheets:

CHECK	RESULT
Caltrans alignment coordinate tables vs the callout fit	median 0.02–0.07 ft on the three Presidio sheets; a second opinion, not better control than the callouts
$L = R \cdot \Delta$ on every curve-table row	68 of 68 hold
Table rows with every cell read	12 rows on R-10434.3 were missing their distance; reader fixed
Record self-consistency	R-10434.1 alignment row L prints northing 2,21,919.67: a drafting error on the record
Same line on adjacent sheets, each fitted alone	≤ 0.17 ft on proven same-line pairs; 12 shared labels on R-10741.2/.3 within 0.19 ft median
Sheet features vs LiDAR (buildings, walls, pavement edges)	none drawn on any of the six sheets; the epoch shift is modelled (NGS HTDP), not measured on the ground

The easements, settled

Tunnel easements 61985-1..4 are one drawn strip, not four figures. Loop 2 chained the 345 single-dash strokes on the `rw_EASE_EXIST_align` layer into eight trains. The strip has a north edge drawn under the R/W line, a south edge parallel to it, and closures at both ends. No stroke on the sheet divides it into four. The parcel bubbles' leaders go to R/W vertex circles, and the printed curve data sits on the R/W line itself.

Measured: the envelope is 48,410 sq ft against a table sum of 70,690. The two temporary construction easements overlap the two tunnel easements rather than tiling the strip. Their individual outlines live in the deeds, not on this sheet.

For the demo: segment 2 shows the two chains that close, the strip envelope with its area, and the four easements queued with this reason. That is the story: the system measures what the sheet draws and says what it cannot prove.

 Strip region of the sheet with the dashed easement trains coloured, the envelope filled, and the four parcel ovals labelled with their table areas

The strip as drawn. Dash trains from the easement layer, the north edge in blue under the R/W line, the envelope filled, the four ovals with their table areas.

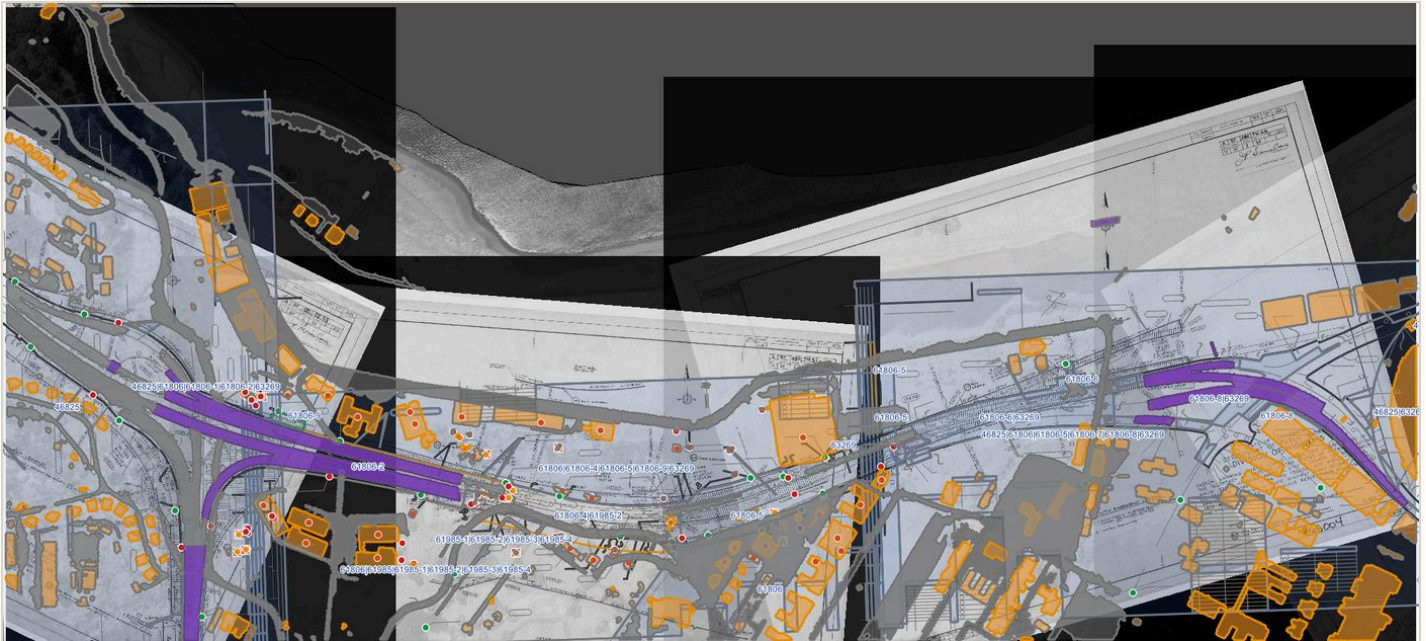
The north set

CHaldenwang's callouts read differently from the Presidio drafter's. Coordinates print as a two-row block, N over a bar over E, with a thin (< 1.0 pt) leader running to a filled arrowhead that sits 10.6–10.9 pt short of a small circle rather than touching it. Civil 3D 2012-style export: no SHX text annotations, no tables, so every block is read by OCR. Three callouts paired on R-10741.1, four on .2, three on .3 (10 of 10 traced) once the leader trace learned two rules: walk across thin weights (below 1.0 pt) instead of stopping at them, and snap to a circle within 12 pt of the arrowhead when no second circle sits within 24 pt.

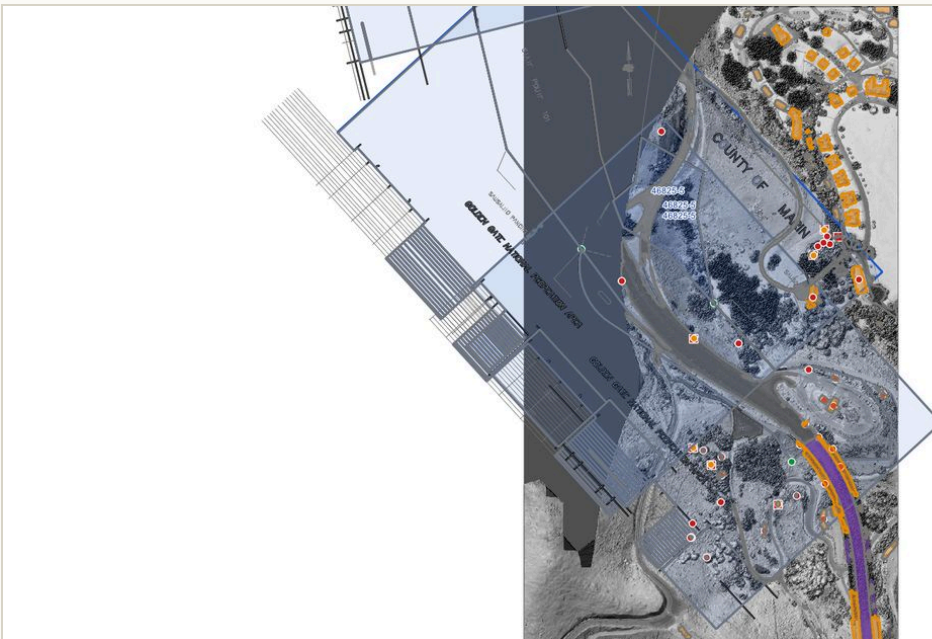
Placement: sheet-derived fits (scale and rotation from the sheet's own bearings and distances, offset from the traced callouts) land at rms 0.04 / 0.05 / 0.03 ft on .1/.2/.3. Checked against Caltrans' package at 9 sheet points: max 0.29 / 0.37 / 0.40 ft. .2 is credible outright; .1 and .3 are weaker fits (3 callouts each) with the rotation confirmed independently by 12 printed bearings apiece. Sheet-derived scale agrees

with the package to 0.011 % and rotation to 0.0051°. The package is now the check on all six sheets and the offset on none.

Coverage: the south tile covers R-10434.1 and .2 only — R-10434.3 is 0 % inside it, its northing starting 5 m past the tile edge (STATE.md's earlier "7 sheets" claim was wrong). The north tile covers R-10741.3 (99.3 %), .2 (59.9 %), .1 (0.3 %). LiDAR lies under four of the six sheets; the bridge and the strait, where neither tile reaches, need USGS 3DEP.



South tile, six.qgz. R-10434.1/.2/.3 linework, parcels and the 1969 record group over the Presidio DEM hillshade and intensity, 27 layers.



North tile, six.qgz. R-10741.1/.2/.3 linework and parcels over the Marin DEM hillshade and intensity, 21 layers.

Next steps

In order, as of 09-26 15:15. Demo preparation leads now: the freeze is 10-03.

01 Slides, Q&A sheet, recorded fallback, rehearsals, freeze NOT STARTED

Non-demo minutes and the scan segment as slides; every number from numbers.md and the bench. The Q&A gains the matchline figure, the gold-set precision and the record error. Record one full run. Two rehearsals inside 25 minutes. No code after 10-03.

02 Coverage, the next causes

Queued drawing labels (33 / 51 / 58 on the three Presidio sheets), several labels claiming one line, centrelines cut by stationing ticks, 42 partly unread tags on R-10434.3. Each leg gated on no pass lost and zero wrong on both gold sets, every new crop viewed.

03 Demo headroom

177 s cold against 180 s. Tag segmentation (51 s of glyph sweeps) is the next cut.

04 The epoch shift on the ground

Needs a tie the LiDAR sees: NGS marks on hard surface, or a sheet that draws structures.

05 Carried

Gap-tile highway surface; C21 and the point-of-tangency class, scoped; scan text-line detection; north row segmentation; R-71.70's 1.29°; the 1950s scans on the north tile.

06 Parked until after 10-05

Scan reading by per-sheet template alphabet. Per-object uncertainty. More drawn-geometry work on 61985-1..4.

What has been built

Forty-eight Python scripts, about 8,000 lines, all under `spike/`. Throwaway by design: they answer "does this work" and feed the demo. Grouped by role; pipeline scripts in the order `demo.py` runs them.

FILE	WHAT IT DOES
Pipeline, in run order	
<code>demo.py</code>	Runs the 21 steps in order offline (<code>--fast</code> , 129 s, and glyph reading when annotation text covers the Routes QGIS steps to QGIS Python. <code>--six</code> adds the five sheets' tails and the QGIS project (640 s, 0 fa
<code>blocks.py</code>	Text blocks from vector strokes: box and angle for piece of lettering, no OC
<code>read_shx.py</code>	Text from AutoCAD SHX annotations, one block annotation. The reader 2016+ sheets. Confidential construction.
<code>frame.py</code>	Where the map is on the map area vs furniture (table block, legends), from the itself.
<code>alphabet.py</code>	A sheet's own alphabet tables: the NO. column in sequence, so its glyphs exemplars. Reads the linear and radial tables.
<code>read_glyphs.py</code>	Fallback reader for sheets without annotations: even matched against the sheet's alphabet. A glyph match
<code>ocr.py</code>	Fallback reader for scan each crop, RapidOCR, per notation.

solve.py + georef.py	One solver over every p control type: N/E callou leaders traced to their p monument notes, grid t Consensus fit to CCS83 : axis redundancy, refuse
tiles.py	Maps a sheet to its LiDA DEM, cache dir under spike/lidar/cache/ the LiDAR tail steps take from it instead of a hard Presidio path.
overlay.py	Linework to NAD83(201 HTDP epoch shift, draw LiDAR intensity image.
parcels.py	Faces from the heavy lir named by parcel labels leaders, checked against table areas.
checks.py	Does what the sheet say what it draws? Each bea distance and arc label a line under the fit; the ex queue.
tags.py + tables.py	Segment tags (L8, C16) i drawing tied to table ro leader arrowhead, else heavy line beside; two c go to the queue. Per-rov
dashes.py	Chains the single-dash s the easement layer into (dash to dash by headin deduplicated by covera checks and parcels see lines as geometry.
traverse.py	The record twin: placed edges chained by share walked by bearing and c closure and area where closes.

<code>lidar/extract.py</code>	Pavement, viaduct deck building polygons from returns, deterministical sheet's frame.
<code>encroach.py</code>	Building-class returns in sheet's own R/W and ea faces; terrain profile over tunnel easements.
<code>export_rasters.py</code>	Intensity ortho and hills GeoTIFFs for QGIS.
<code>objects.py</code>	The object record: every thing with sheet, printed measured values, residue status, adjudication slot
<code>exceptions_page.py</code>	One offline HTML page: queued item with crop, printed vs drawn, citatic resolution field. Scan's o the same page.
<code>qgis_project.py</code>	The screen: PyQGIS built project with every layer ordered, re-reads it, ren opening view as proof. builds two tile groups (2 layers) and renders one tile.
<code>figures.py + slide_numbers.py</code>	One figure per demo sequence rendered from the project number the slides quoted run's files.

Measurement and gates

<code>bench.py</code>	The gate instrument: run chain on a sheet set and summarises pass, fail, w tags, faces, traverse, take sheet, plus frame / bloc bearings_parsed / distances_parsed over t
-----------------------	---

	sheet set. Every loop leg accepted or rejected on
gt.py, gt_scan.py	Hand-keyed truth: 227 t on R-10434.2, self-check R-Δ; 34 boxes on the 196
sheets.py, batch.py	Generalisation runs: the chain over every vector disk and over unrelated maps from the public in
leg4_crops.py, legB_arcs.py, leg5_61985.py, checks_debug.py	Gate evidence and diag random passing distanc and every passing arc re their lines; the four ease walked as chains and th envelope measured; the line per label drawn.
Scans (most of the archive)	
scan_read.py, scan_readers.py, scan_vlm.py	Detect text boxes on a 3 render; RapidOCR and a vision model (qwen2.5v against keyed truth. No values in any prompt.
scan_consensus.py, scan_georef.py	Accept a read only wher readers agree or structu it; fit callouts through th solver, refused unless sc matches the grid labels.
scan_on_tile.py	The four 1969 R-65 scan georeferenced by Caltra packages, HTDP-shifted LiDAR tile as the "1969 r QGIS group.
LiDAR questions answered once	
lidar/q1..q5	Flight date from GPS tin 09-27); record on terrair epoch offset by real NG! encroachment; flightlin QA (RMSE 0.02–0.04 m c

Documents

STATE.md, ROADMAP.md, spike/LOOP.md

Resume point with next every known trap; the 1 and the Q&A sheet; the loop's legs, gates and o

spike/out/loop_report.md, legC_misses.md, numbers.md, exceptions.html

The overnight report; th attribution; the slide nu queue.

Findings worth carrying forward. Annotation text is the reader for Civil 3D 2016+ sheets; the glyph alphabet stays for 2012-era sheets. CAD layers did not help association once reads were exact; they still separate furniture from map. The exception page's wrong-line group is the fastest diagnostic there is: render the measured line, look at twelve crops, fix the biggest cause, repeat. Two generic readers agreeing on hand lettering is a floor, not a method.

```
SOURCES: STATE.MD, ROADMAP.MD, SPIKE/LOOP.MD,
SPIKE/OUT/LOOP_REPORT.MD, SPIKE/OUT/LEGC_MISSES.MD,
SPIKE/OUT/NUMBERS.MD, SPIKE/OUT/BENCH.CSV (ROW "LOOP19-4"
FOR THE RECONSTRUCTABILITY AND GOLD TABLE; ROW "LEGI" FOR
THE DEMO-SHEET SCORECARD BELOW IT; GOLD SETS
SPIKE/GOLD/*_ASSOC.JSON; SET-LEVEL HEADLINE
SPIKE/OUT_RECON/SET_RECON.JSON; PER-SHEET RECONSTRUCTION
SPIKE/OUT_RECON/*.JSON AND FIGURES; COVERAGE
SPIKE/OUT_COVERAGE/ATTRIBUTION.MD; MATCHLINES
SPIKE/OUT_GROUND/MATCHLINE.MD; RECORD CHECKS
SPIKE/OUT_RECORD/SUMMARY.MD; SCAN NUMBERS FROM
SPIKE/GT_SCAN.PY AGAINST READ_SCAN.JSON AND
READ_ALPHABET.JSON) · EVERY NUMBER RE-MEASURABLE WITH
PYTHON SPIKE/BENCH.PY FINAL --TABLES --TAGS --PARCELS --TRAVERSE
PRESIDIO R10434_1 R10434_3 R10741_1 R10741_2 R10741_3 R71_71
R10258 R71_70
```